WTP DISINFECTION

UV, Chlorine & AS2927 Owner Compliance Primer

Purpose

This resource is useful enough to help asset owners identify issues, keep useful records, and decide when to escalate — it is not a repair guide, design guide, or compliance certificate.

A plain-language primer on dual disinfection at a water treatment plant — UV, chlorine, contact time, and the AS2927 chlorine gas safety obligations every plant owner must meet.

Important Scope Note

This guide is general information only. It is not design advice, certification, or a substitute for a process audit, HAZOP, or a formal AS2927 assessment.

Use it to understand what good disinfection looks like at owner level, what AS2927 expects of you as an owner, and what compliance evidence to demand. Do not use it to size UV reactors, set chlorine doses, design ventilation, change alarm setpoints, or interpret AS2927 clauses for a specific site.

What This Guide Helps You Do

- Understand why dual disinfection (UV + chlorine) is the modern standard for surface water.
- Interpret UV and chlorine specifications at owner level.
- Understand Ct (concentration × time) and why it is calculated at peak flow.
- Know what AS2927 requires you to provide as a chlorine gas plant owner.
- Recognise the critical control points an owner should be able to name in a meeting.

What This Guide Does Not Cover

- UV reactor sizing or validation procedures.
- Chlorine system design or carrier-water calculations.
- Clause-by-clause interpretation of AS2927.
- HAZOP or risk assessment procedure.
- Certification of compliance for a specific site.
- Design dose selection for any specific source water.

Why Dual Disinfection Is the Standard

No single disinfection technology covers the full pathogen spectrum. The two work together because they cover each other's weaknesses.

Pathogen group	UV contribution	Chlorine contribution
Bacteria	Strong	Strong
Viruses	Limited	Strong
Protozoa (including Cryptosporidium and Giardia)	Strong	Effectively none

UV inactivates protozoan cysts that chlorine cannot touch. Chlorine inactivates viruses that UV does not address efficiently — and provides a residual disinfectant in the distribution network that UV cannot, since UV has no carry-through effect once water leaves the reactor.

A modern WTP treating Category 3 or Category 4 source water relies on both, in series. Treating them as alternatives is a common design and procurement error.

WTP DISINFECTION

UV Disinfection — What Owners Need to Understand

UV disinfection exposes microorganisms to ultraviolet light at the 254 nm wavelength, damaging their DNA or RNA so they cannot reproduce.

Concept	What it means at owner level
UV dose (mJ/cm²)	The product of UV intensity and exposure time. Validated reactors deliver a target dose at a given flow and water quality.
UV transmittance (UVT) at 254 nm	The proportion of UV that passes through the water. Lower UVT means less effective disinfection.
Validated LRV	UV reactors are validated against recognised protocols (such as USEPA UVDGM or DVGW). The validation defines what dose, what UVT, what flow range.
Warm-up period	Lamps take several minutes to reach full intensity from a cold start. Treated water cannot leave the plant during warm-up. This is a Critical Control Point.
End-of-lamp-life	Lamp output declines with hours. Replacement is scheduled, not reactive.
Fouling factor	Quartz sleeves foul over time, reducing transmitted intensity. Wiping systems and cleaning intervals are part of the design.
Redundancy	Duty / standby reactor configuration is the modern standard. A single UV reactor is a single point of failure for the entire plant.

Chlorine Disinfection — What Owners Need to Understand

Chlorine can be dosed as gas (high-purity, low cost per kg, more demanding to handle) or as sodium hypochlorite (lower hazard, higher cost, decay over time). Both deliver chlorine residual into distribution; the choice depends on site, scale, supply chain, and operator capability.

For gas systems, the key concepts at owner level:

- **Vacuum-feed chlorinators** — gas is drawn under vacuum, not pushed under pressure. A leak in the gas line collapses the vacuum and stops dosing rather than releasing gas.
- **Carrier water** — the gas dissolves into a treated-water carrier stream before injection into the main flow. Carrier water reliability is part of the system design.
- **Residual analysers** — free chlorine analysers at the carrier-water injection point and at the chlorine contact tank outlet confirm that dosing is performing.
- **Cylinder configuration** — cylinder banks are sized for the relevant remote-site resupply window, with duty / standby manifolds and automatic regulators.

Ct (Concentration × Time) — The Owner-Level Concept

Ct is the product of free chlorine concentration (mg/L) and contact time (minutes). The ADWG sets minimum Ct values for effective viral and bacterial inactivation in distribution. The owner-level points:

- The chlorine contact tank (CCT) volume and its **baffle factor** together determine the effective contact time for a given flow. A poorly baffled tank gives much less effective contact than its volume suggests.
- Ct must be calculated at **peak flow**, not average flow. A plant that meets Ct at average flow but fails at peak is non-compliant during the hours that matter most.
- "Validated" Ct means the calculation has been confirmed by testing or recognised methodology — not assumed from nameplate volume.

AS2927 Chlorine Gas Safety — Owner Obligations

AS2927 governs the storage and handling of chlorine gas at water and wastewater facilities. The owner's obligations sit alongside the designer's and the contractor's — and survive long after handover.

At owner level, AS2927 expects you to provide and maintain:

- **Gas leak detection** with auto-shutdown of the chlorination system on leak alarm.
- **Forced ventilation** of the gas storage and dosing rooms.
- **Self-closing doors** and restricted access to the gas areas.

WTP DISINFECTION

- **Breathing apparatus** appropriate to the cylinder size on site, with documented inspection.
- **Emergency shower and eyewash** with regular testing.
- **Hazardous area classification** documented and visible.
- **Trained personnel** — only trained operators may enter and work in the gas areas; documented competency.
- **Drilled emergency procedures** — gas leak, evacuation, no-entry-during-alarm protocols, emergency services contact.

Compliance is demonstrated by documentation: the AS2927 risk assessment, the design-stage compliance report, ventilation commissioning records, gas leak detector test logs, breathing apparatus inspection records, and operator training records. An owner who cannot produce these on request is exposed.

Critical Control Points an Owner Should Be Able to Name

A WTP operator should be able to name and demonstrate every Critical Control Point. An owner should be able to name them in a meeting:

- UV In-Service interlock — no treated water leaves the plant unless UV is confirmed at validated dose.
- Free chlorine residual at CCT outlet — minimum residual must be confirmed before transfer to reservoirs.
- Gas leak detection auto-shutdown — chlorination stops automatically on any leak alarm.
- CCT contact time achieved at peak flow — Ct is met across the operating range.
- Alarm escalation pathway — every critical alarm has a documented escalation step.

Common Disinfection Compliance Failures

- UV and chlorine treated as alternatives instead of complementary barriers.
- UV reactor present but no validation evidence on file — performance cannot be claimed.
- CCT undersized or unbaffled at peak flow — Ct fails at the hours that matter.
- AS2927 documentation incomplete or out of date — the design-stage report exists but operational records do not.
- Free chlorine residual not measured at distribution entry — only at the CCT outlet.

Owner Decision Prompts

Questions to ask the designer, contractor, or operator before audit, handover, or re-procurement:

- What LRVs do UV and chlorine each contribute, and what is the total against ADWG requirements?
- What is the design Ct at peak flow, and what baffle factor is assumed for the CCT?
- Is the UV reactor validated, and against what protocol — at what dose, what UVT, what flow range?
- Where is the AS2927 compliance audit, the design-stage report, and the current operational records?
- How is gas leak detection interlocked with auto-shutdown, and when was it last tested?
- Where is free chlorine measured — only at the CCT outlet, or also at distribution entry?

Next Step

Disinfection compliance is the part of WTP operation an auditor, regulator, or insurer will look at first. The hardest part is rarely the equipment — it is the documentation, the validation evidence, and the AS2927 records that need to stand up to a third party.

PC Water Solution reviews dual-disinfection systems against ADWG, AS2927, and AS1851 expectations. We cover UV validation, Ct calculations, gas safety obligations, and the documentation an auditor or insurer wants to see — and tell you where your compliance position is defensible and where it is exposed.

Book a WTP disinfection compliance review — call 1300 029 804, email info@pcwater.com.au, or visit pcwater.com.au.